C-STS color clustering: key results

Fine-tuned Qwen3-Embedding-8B, with the original color condition included

What the earlier numbers mean. The single-color validation subset contains 13 color labels; 12 groups gave the best typical separation score. The test subset contains 11 color labels and favors 11 groups. Across all 1,017 inputs, the selected result is 20 groups.

Winning encoder setup

Checkpoint: proposition_final.pt | **LoRA:** rank 32, alpha 32, dropout 0.05
Embeddings: final non-padding end token, 4,096 dimensions, unit-length normalized; 160-token limit.
Clustering: k-means++ / Lloyd; 20 random starts per cluster count; at most 500 iterations. Existing frozen vectors are reused.

Cluster-count winners for this encoder

Input group	Inputs	Colors*	Chosen K	Median score	Single-run K	Highest score
Validation / one color	65	13	12	0.8012	11	0.8384
Test / one color	119	11	11	0.8561	10	0.8730
All splits / one color	497	17	12	0.8156	11	0.8245
All splits / all inputs	1017	17	20	0.5675	21	0.5934
Validation / all inputs	137	13	25	0.5626	24	0.5996
Test / all inputs	239	11	21	0.5945	16	0.6323
Training / one color	313	17	12	0.7872	12	0.8109
Training / all inputs	641	17	16	0.5763	19	0.5932

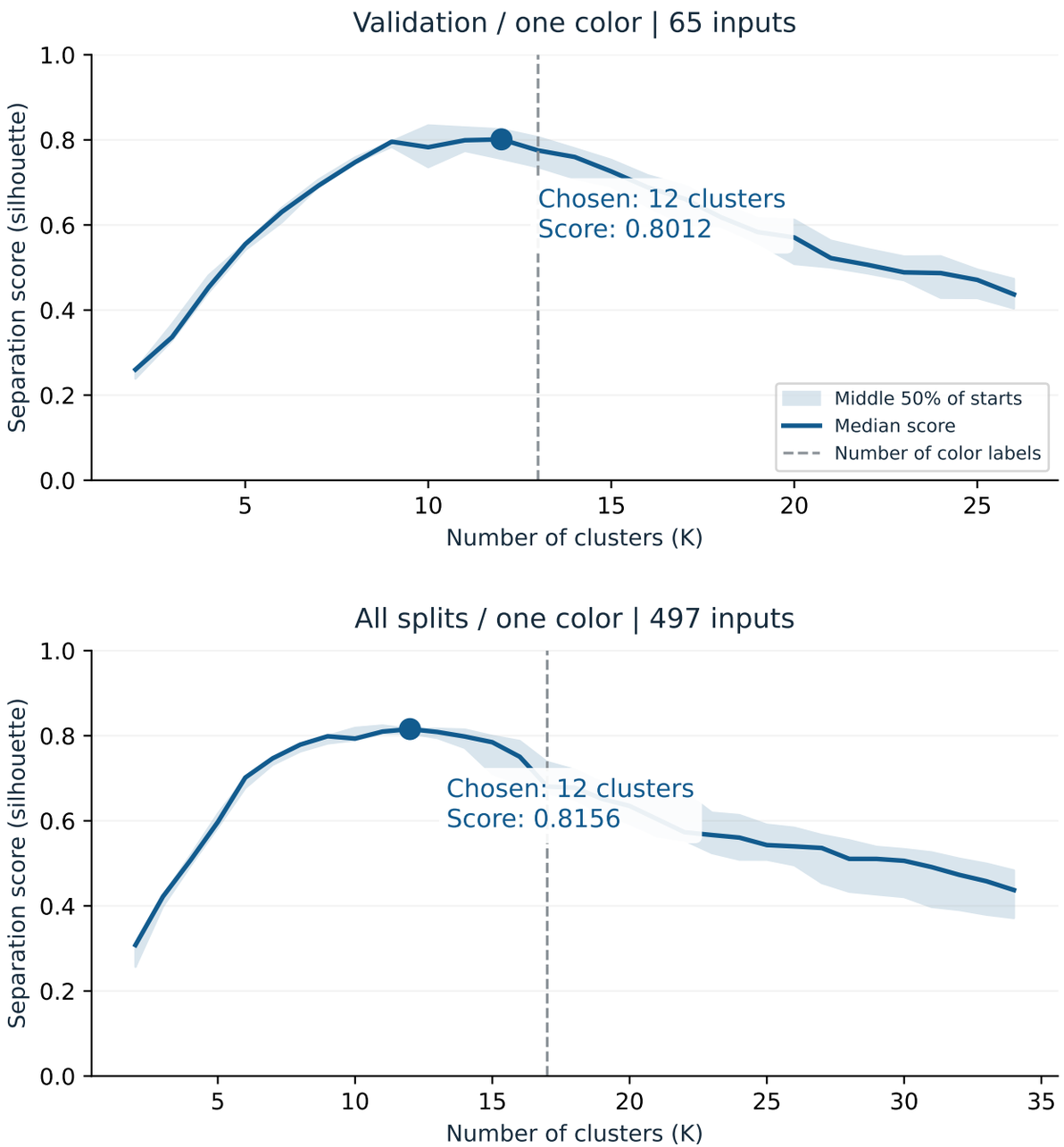
Chosen K: highest median silhouette score over 20 starts. The median rewards a result that repeats across starting points. **Highest single run:** the best individual fit across all counts and starts. The cluster pages use the best start at the chosen K.

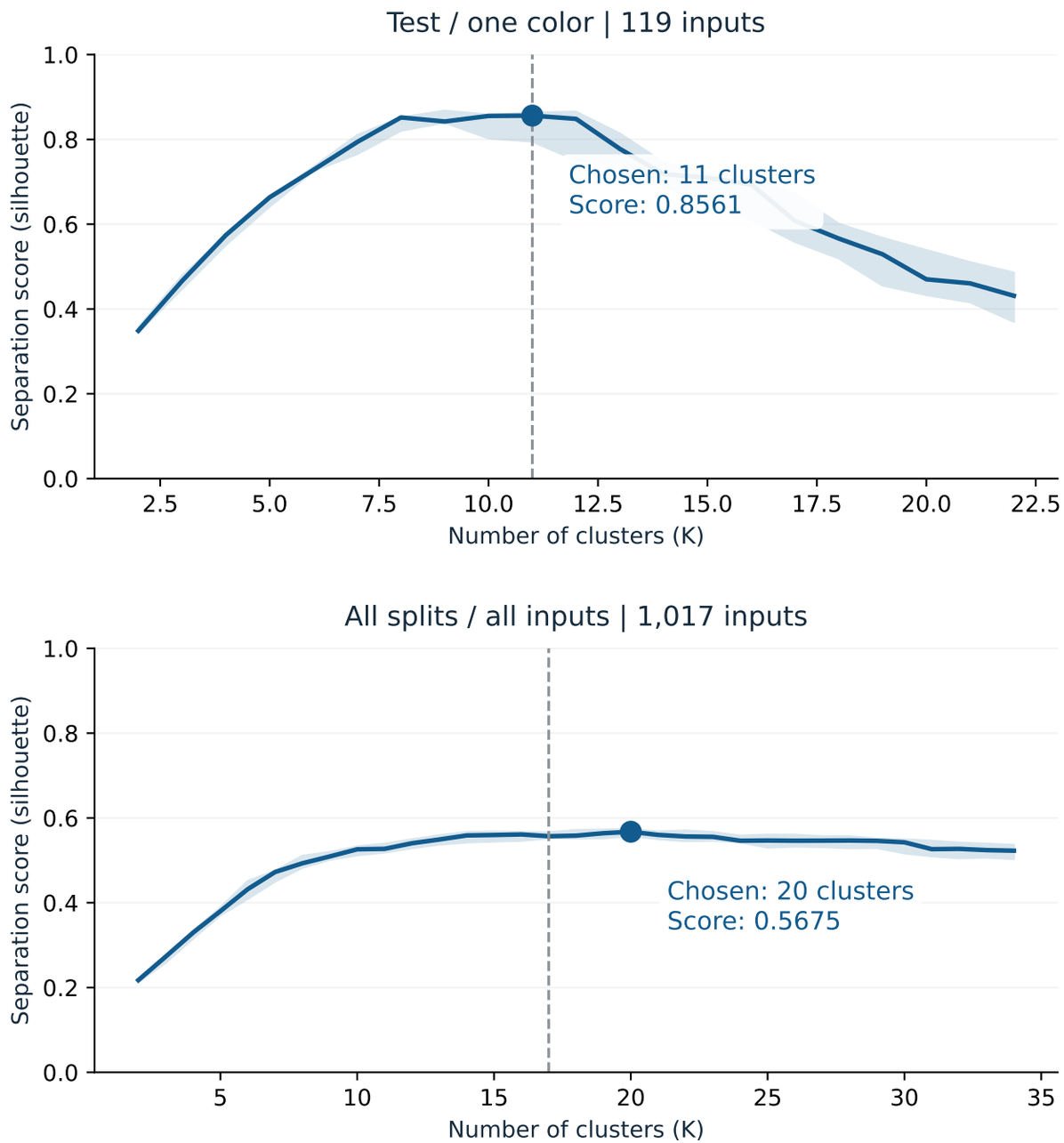
* Colors counted among clearly resolved, single-color inputs. Other inputs stay in the all-inputs analyses. Labels are provisional.

Scores range from -1 to 1; higher means groups are tighter and further apart. No encoder training or clustering is repeated for this pack.

Which cluster count gives the clearest groups?

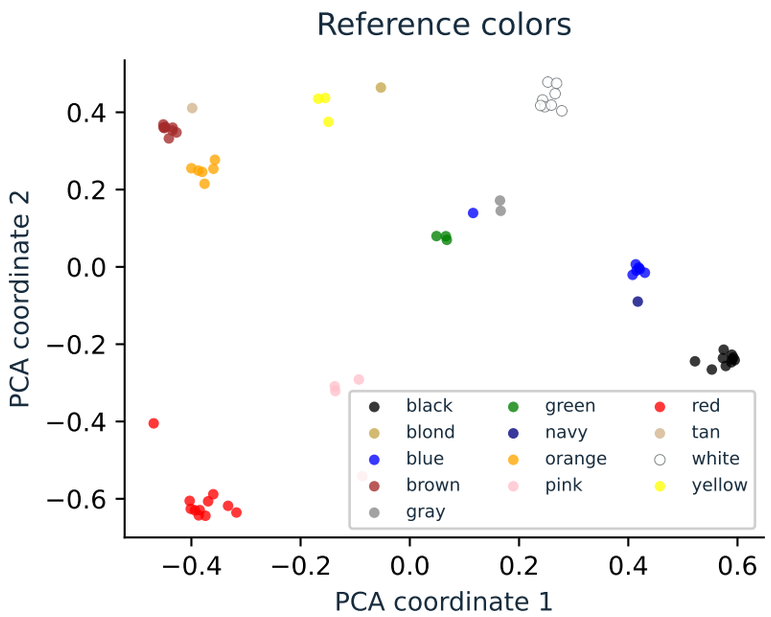
Each curve uses the winning fine-tuned encoder. Blue dots mark the highest median score across 20 starts.

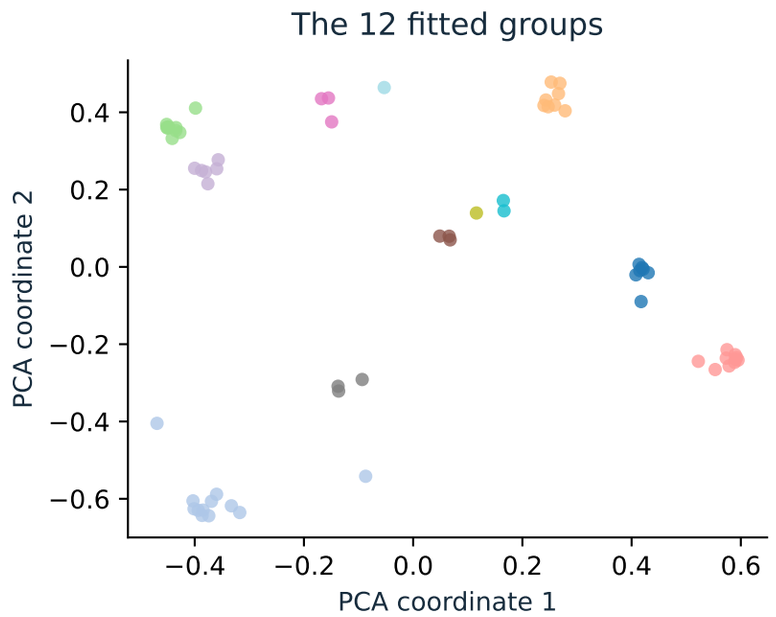


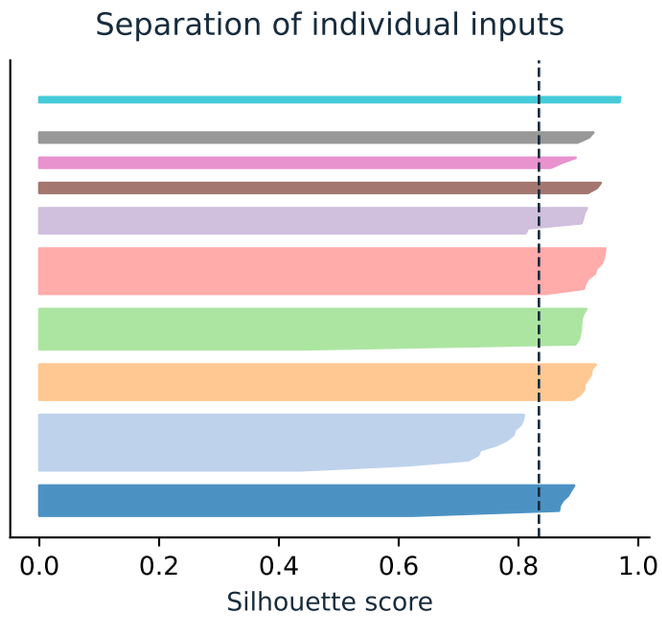


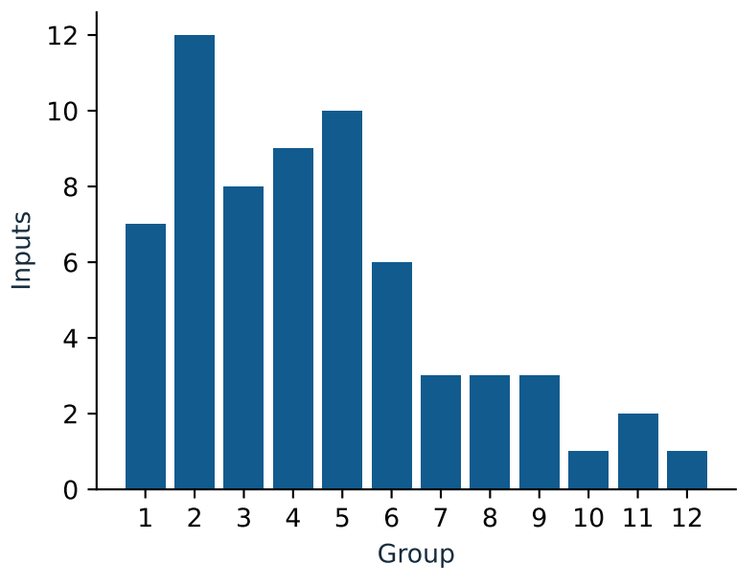
Validation / one color: 12 clusters

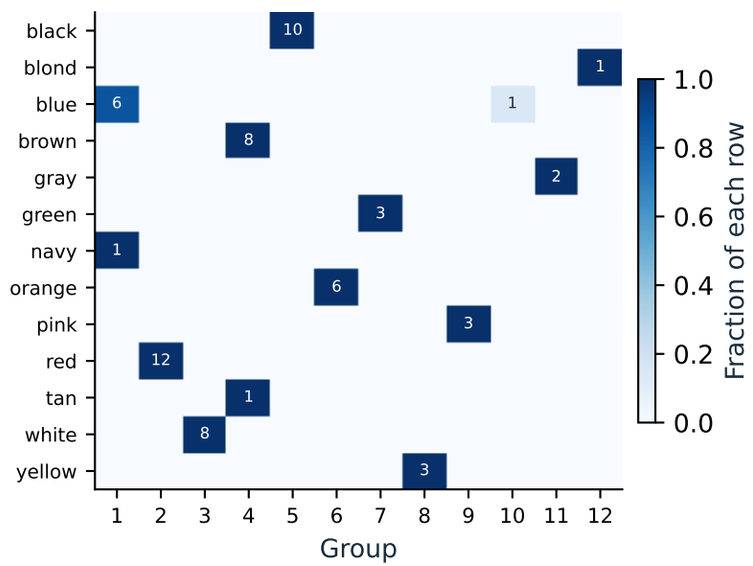
65 inputs | 13 known colors | Best start at chosen K: seed 2 | Silhouette 0.8341 | Color agreement (ARI) 0.9486

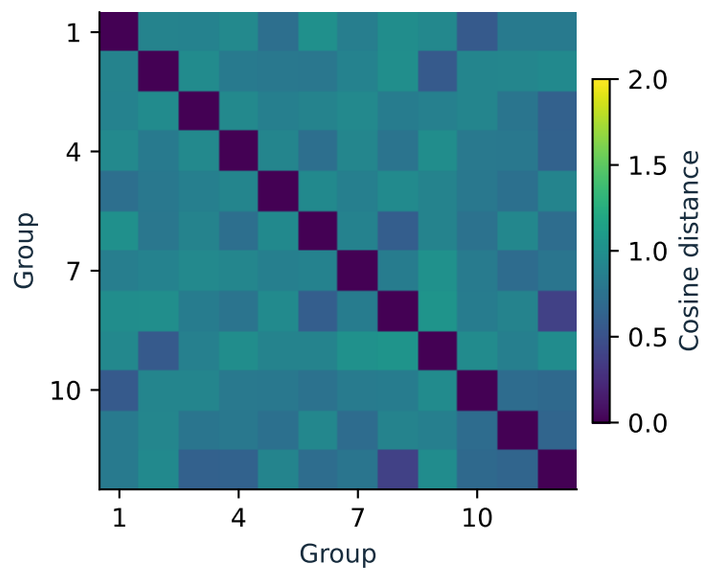






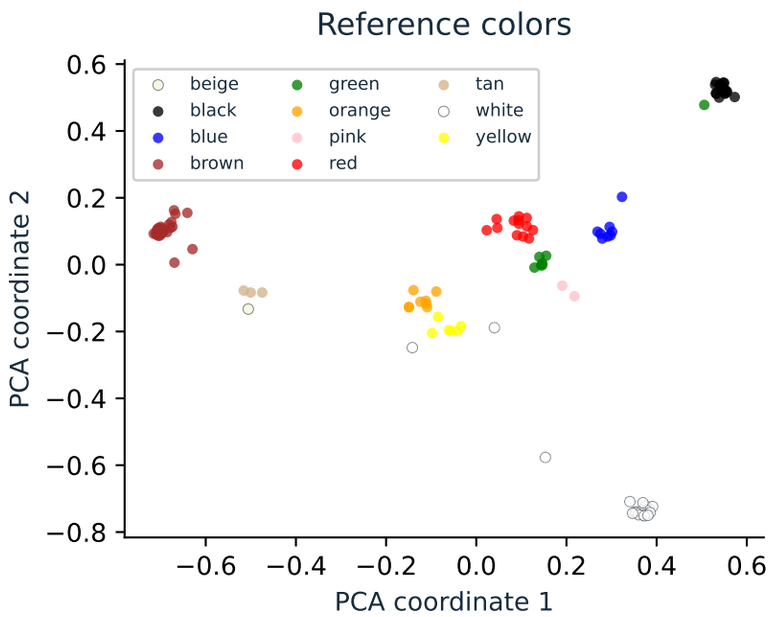


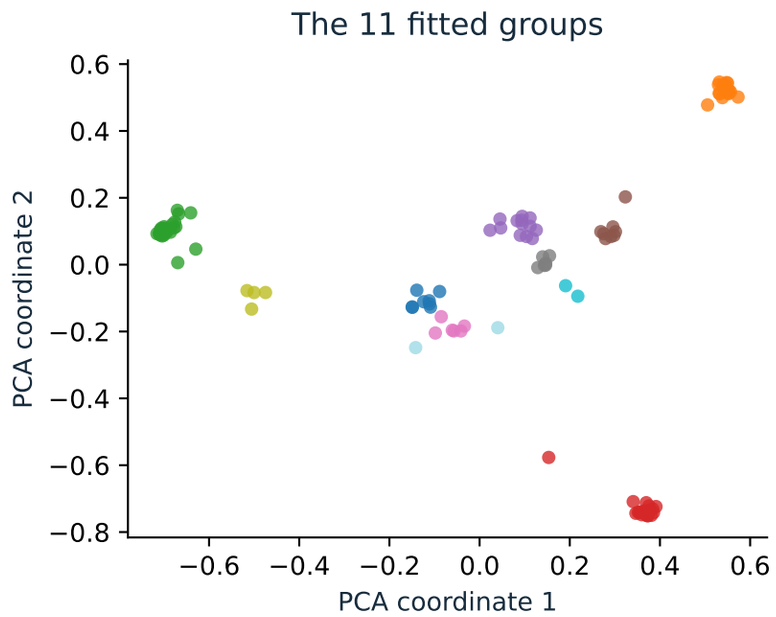


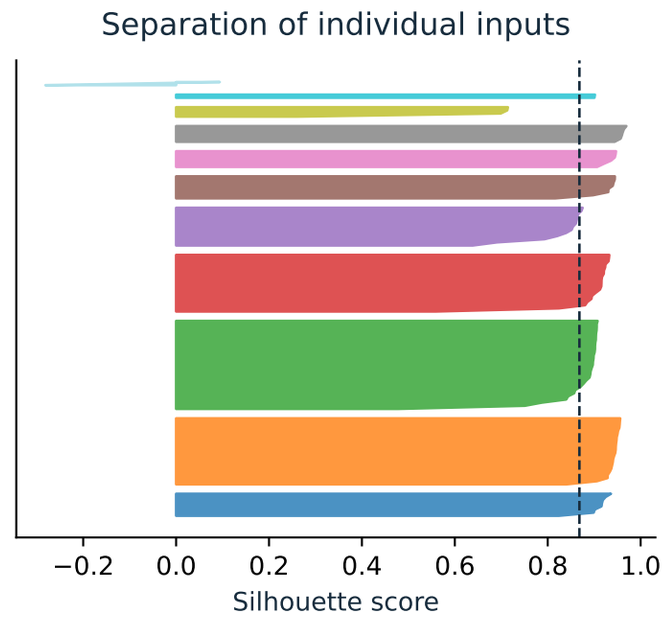


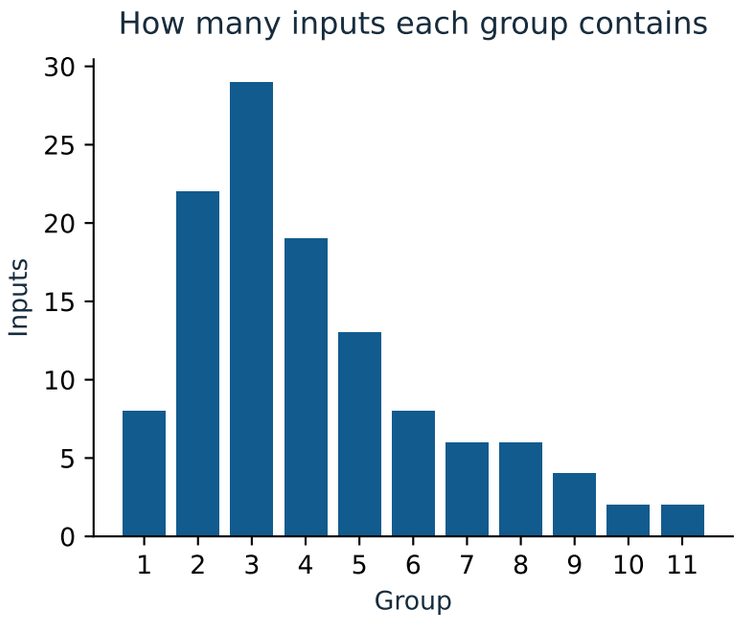
Test / one color: 11 clusters

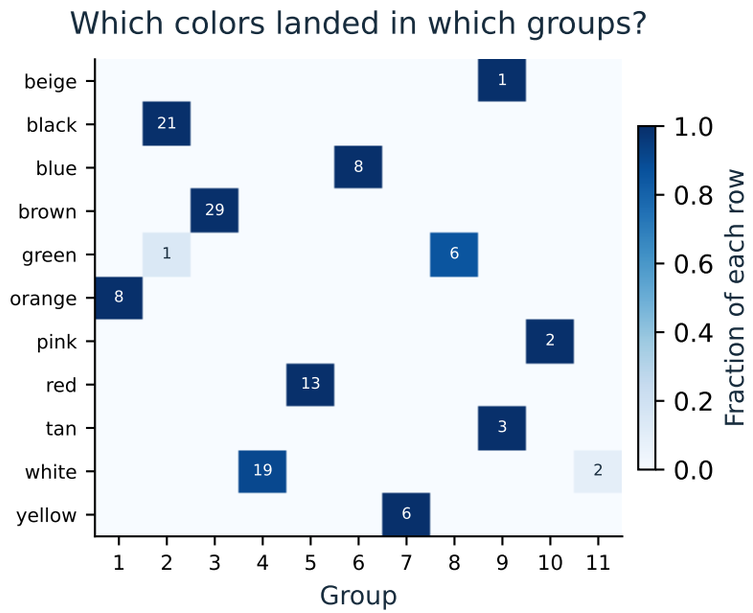
119 inputs | 11 known colors | Best start at chosen K: seed 0 | Silhouette 0.8682 | Color agreement (ARI) 0.9600

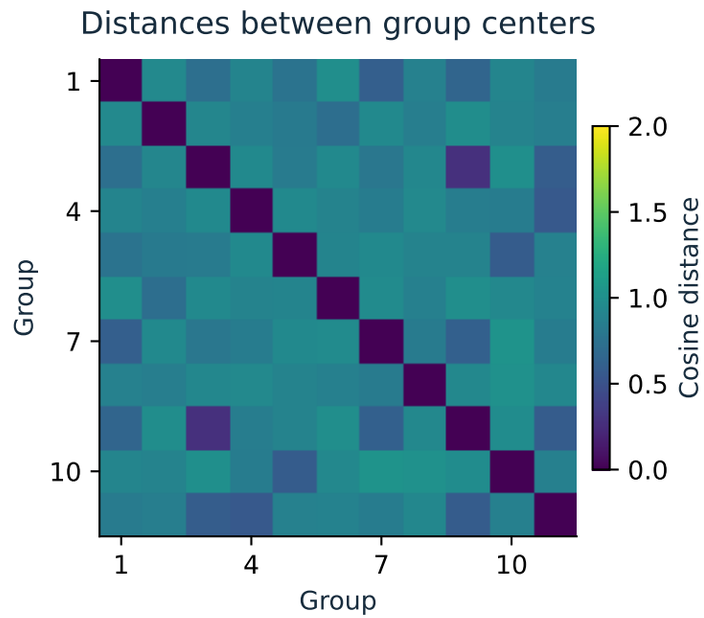






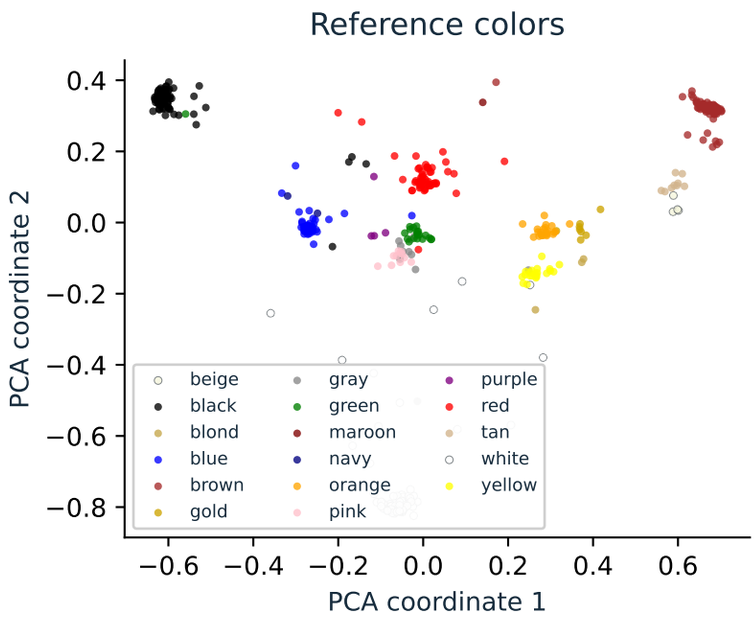


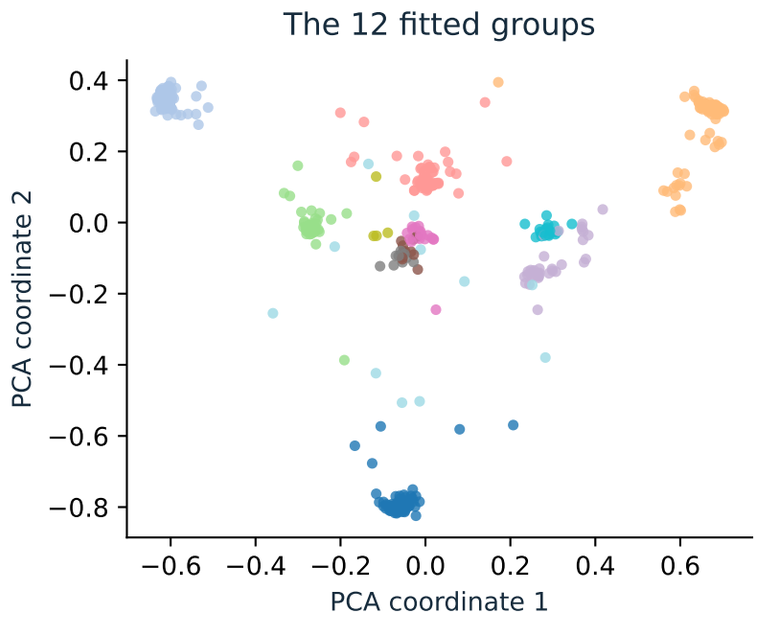


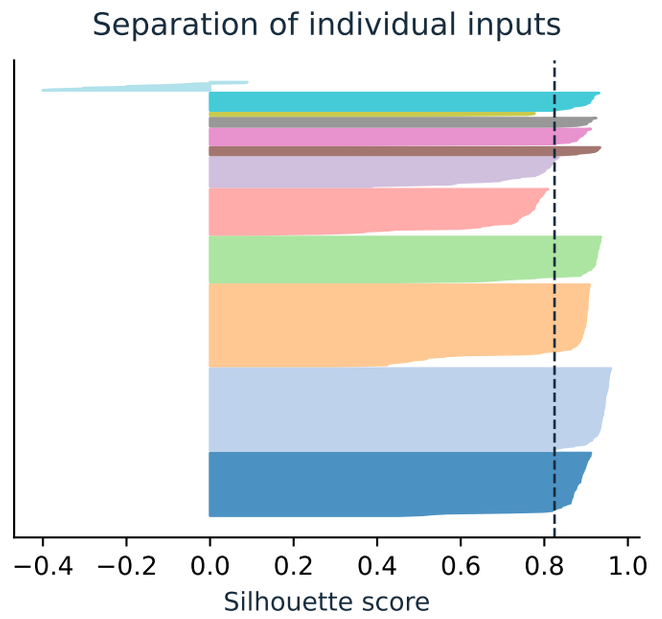


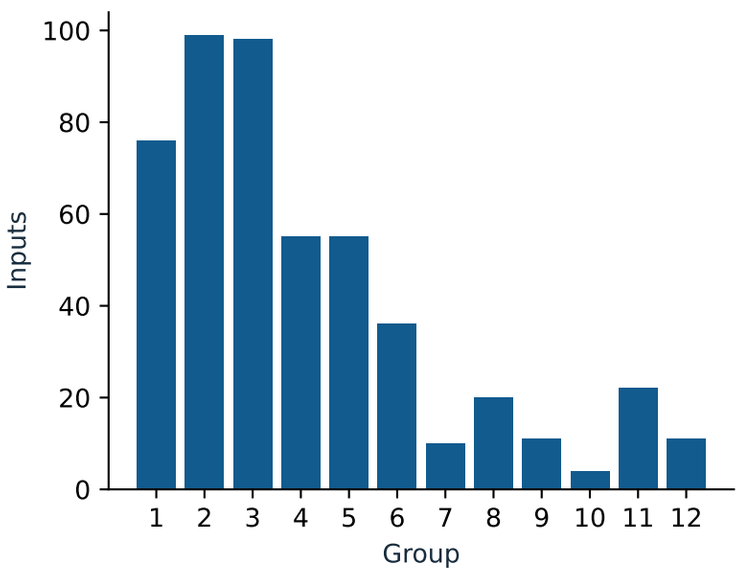
All splits / one color: 12 clusters

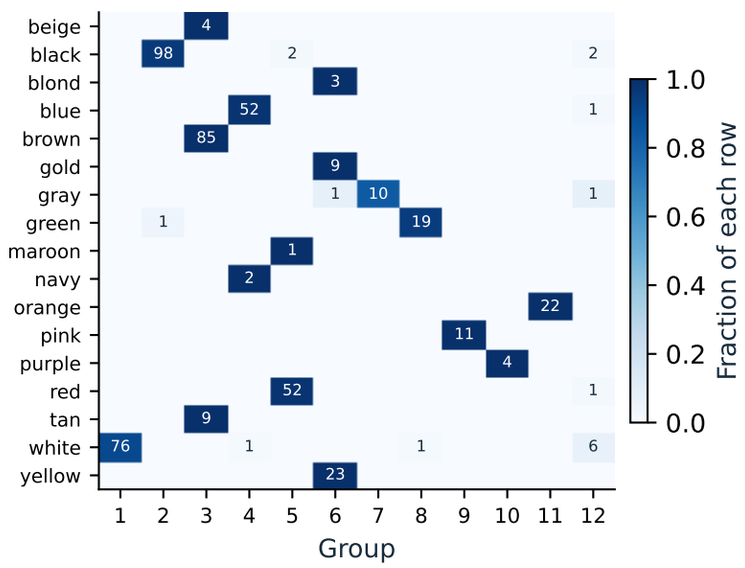
497 inputs | 17 known colors | Best start at chosen K: seed 17 | Silhouette 0.8244 | Color agreement (ARI) 0.8897

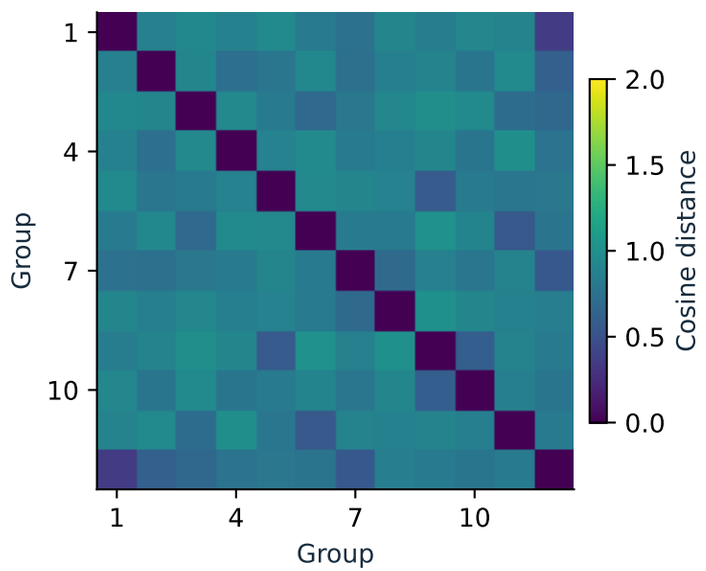








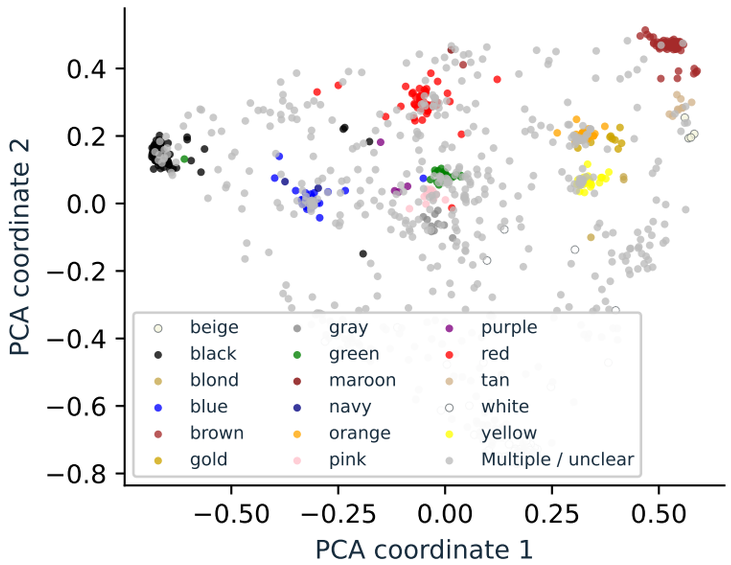




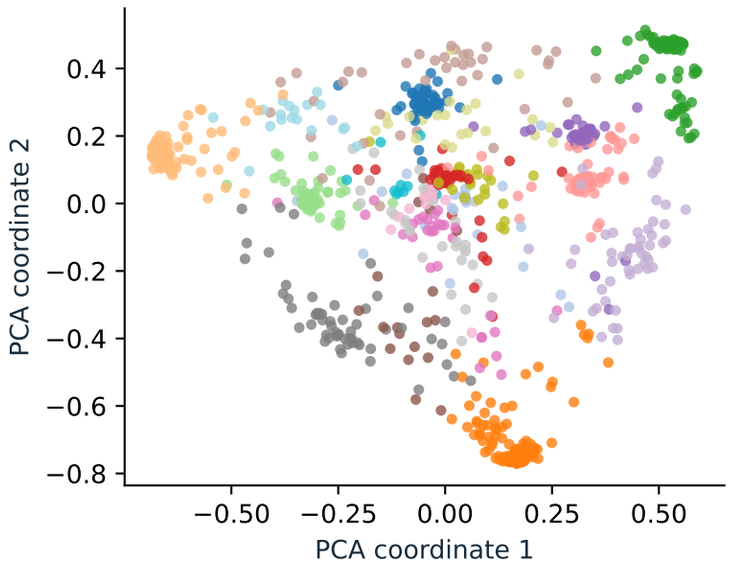
All splits / all inputs: 20 clusters

1,017 inputs | 17 known colors | Best start at chosen K: seed 7 | Silhouette 0.5896 | Color agreement (ARI) 0.8854

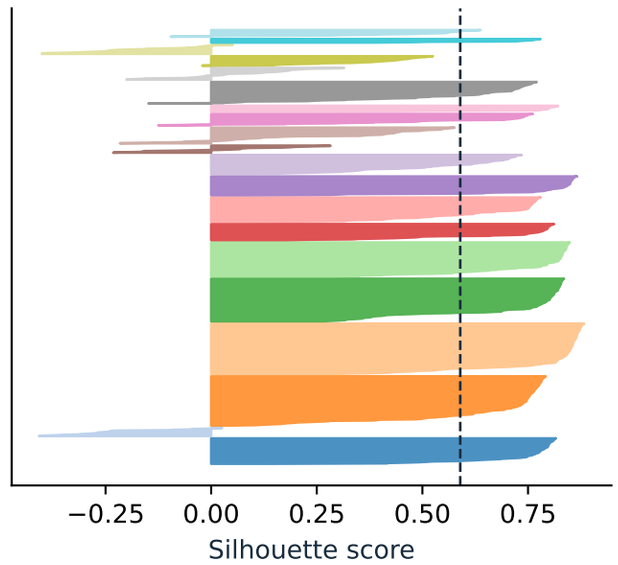
Reference colors



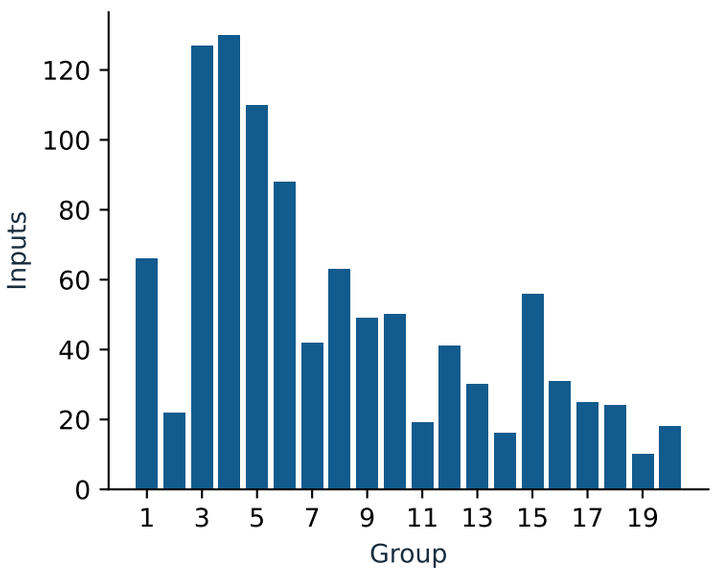
The 20 fitted groups



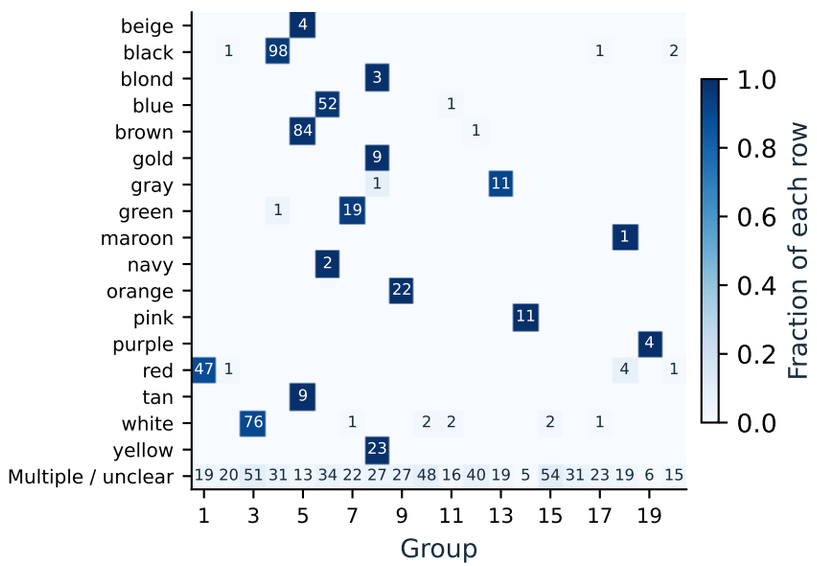
Separation of individual inputs



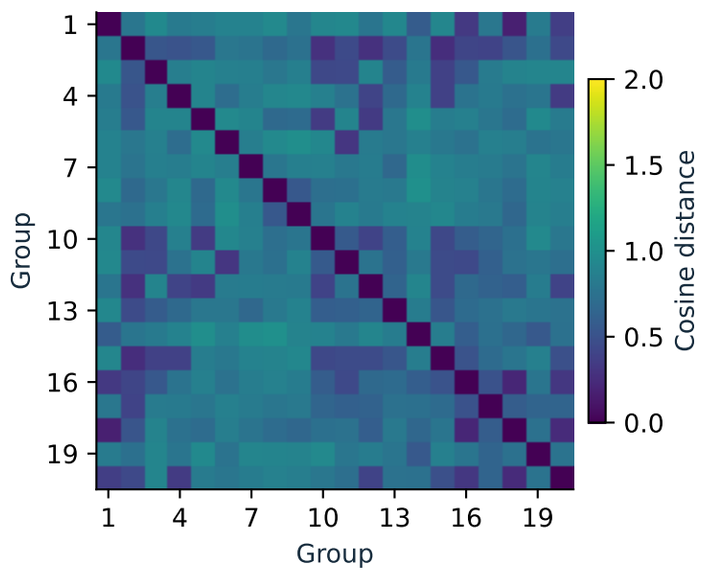
How many inputs each group contains



Which colors landed in which groups?



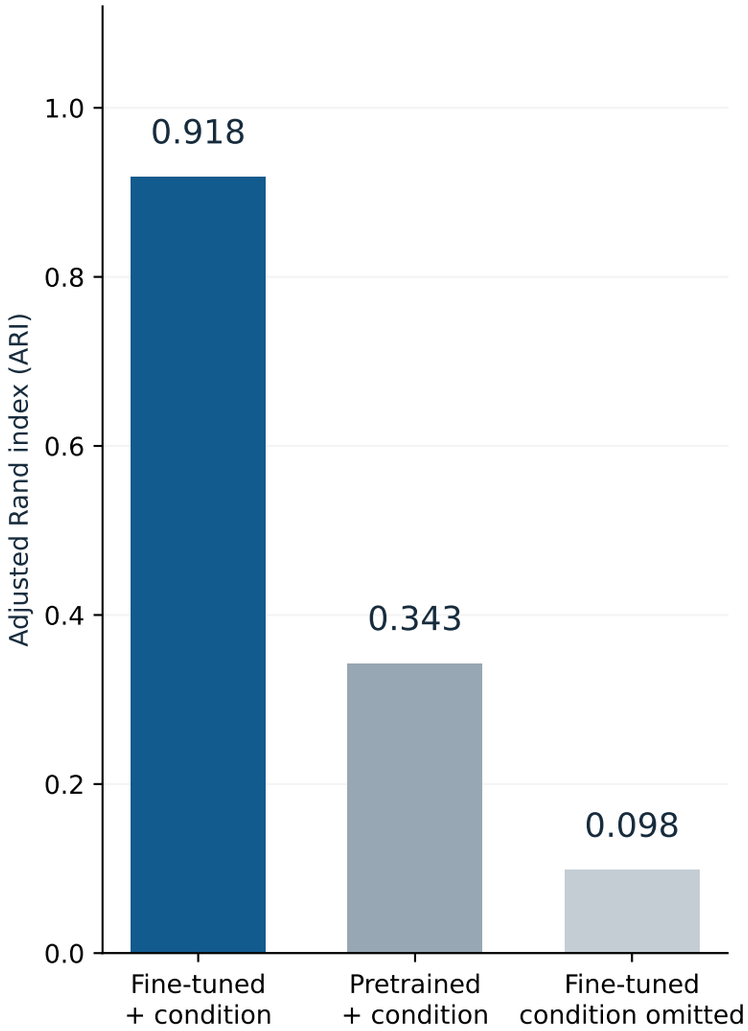
Distances between group centers



Fine-tuned Qwen3-Embedding-8B and the two controls

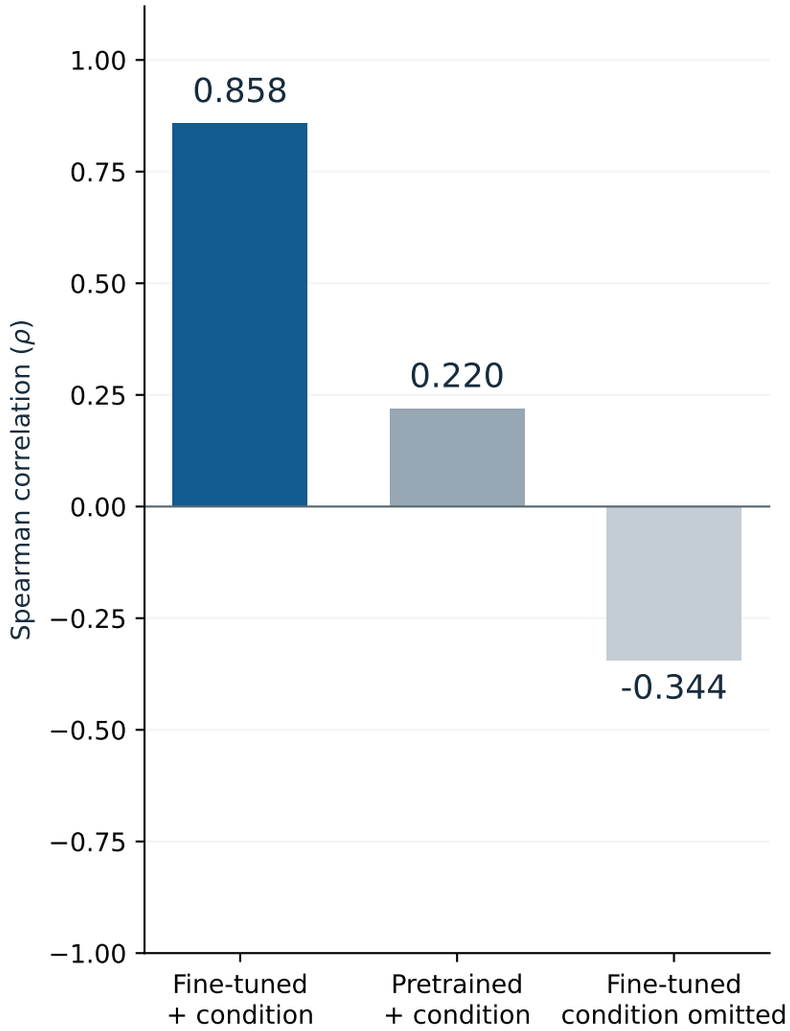
Left and middle: validation metrics. Right: exploratory condition-switch probes drawn from the dataset.

Color-label agreement at 13 clusters



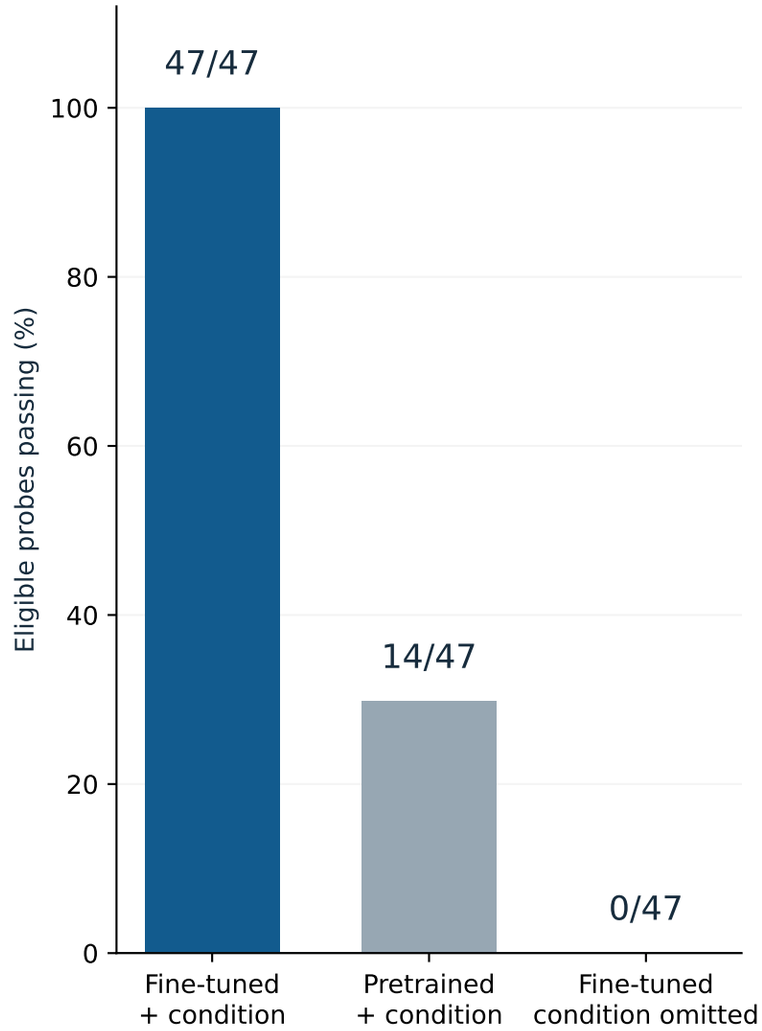
65 single-color validation inputs
K=13 for every model; median of 20 starts

Cosine similarity vs audited ratings



74 color-related validation pairs
Raw cosine compared with audited ratings

Correct color under both conditions



47 eligible of 50 source-derived probes
Mixed splits; reference colors from other families